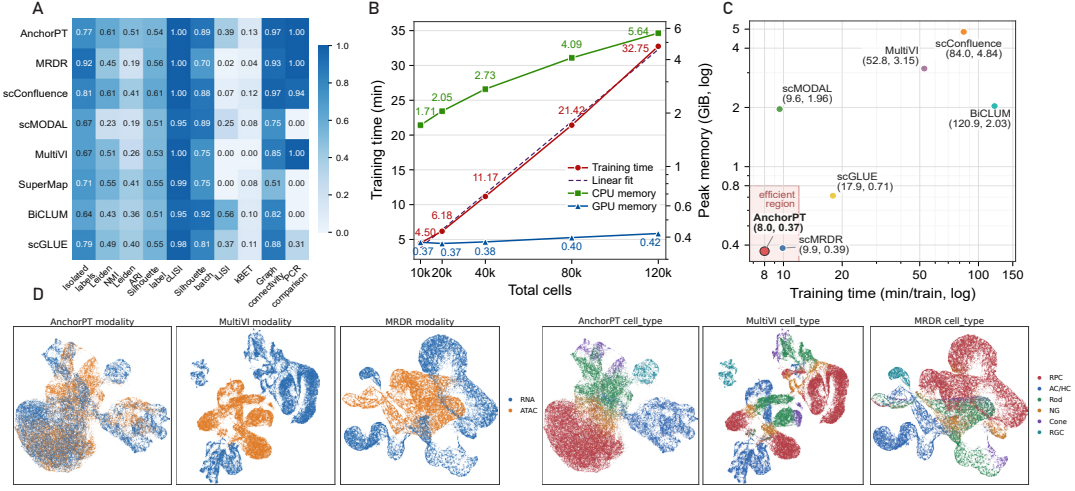

A
AnchorPT 0.77 0.61 0.51 0.54 1.00 0.89 0.39 0.13 0.97 1.00
MRDR 0.92 0.45 0.19 0.56 1.00 0.70 0.02 0.04 0.93 1.00
scConfluence 0.81 0.61 0.41 0.61 1.00 0.88 0.07 0.12 0.97 0.94
scMODAL 0.67 0.23 0.19 0.51 0.95 0.89 0.25 0.08 0.75 0.00
MultiVI 0.67 0.51 0.26 0.53 1.00 0.75 0.00 0.00 0.85 1.00
SuperMap 0.71 0.55 0.41 0.55 0.99 0.75 0.00 0.08 0.51 0.00
BiCLUM 0.64 0.43 0.36 0.51 0.95 0.92 0.56 0.10 0.82 0.00
scGLUE 0.79 0.49 0.40 0.55 0.98 0.81 0.37 0.11 0.88 0.31
Isolated labels
Leiden NMI
Leiden ARI
Silhouette label
kBET
Graph connectivity
PCR comparison
1.0
0.8
0.6
0.4
0.2
0.0
B
Training time (min)
Total cells
10k 20k 40k 80k 120k
Peak memory (GiB, log)
Training time
Linear fit
CPU memory
GPU memory
1.71
2.05
2.73
4.09
5.64
4.50
6.18
11.17
21.42
32.75
0.37
0.37
0.38
0.40
0.42
C
Training time (min/train, log)
MultiVI (52.8, 3.15)
scConfluence (84.0, 4.84)
scMODAL (9.6, 1.96)
BiCLUM (120.9, 2.03)
scGLUE (17.9, 0.71)
efficient region
AnchorPT (8.0, 0.37)
scMRDR (9.9, 0.39)
D
AnchorPT modality
MultiVI modality
MRDR modality
RNA
ATAC
AnchorPT cell_type
MultiVI cell_type
MRDR cell_type
RPC
AC/HC
Rod
NG
Cone
RGC